UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	NOVEMBER
TIME	2026

INSTRUCTIONS TO MARKERS

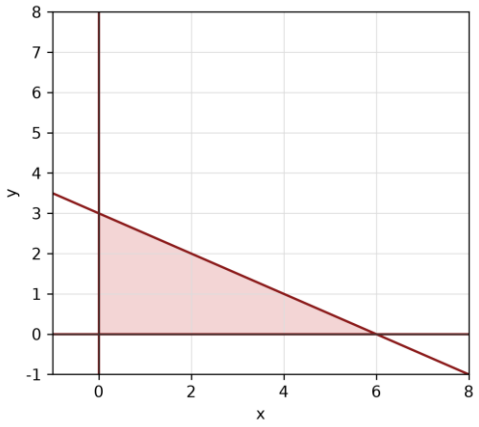
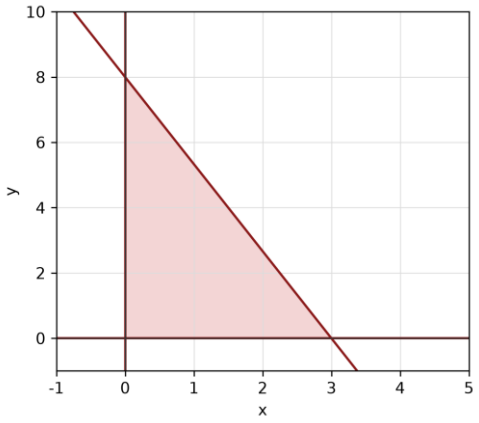
1. This marking scheme accompanies the MAT133 mock paper (November 2026).
2. Answers and marking points are for local study use only.

FOR MARKERS' USE ONLY

This marking scheme consists of 8 printed pages

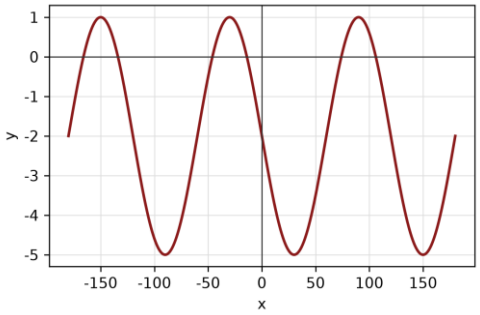
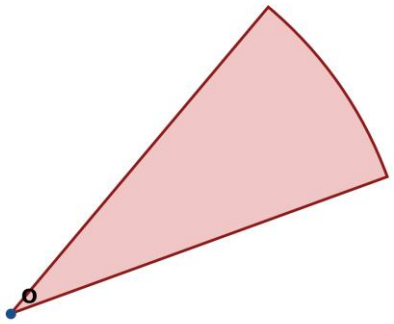
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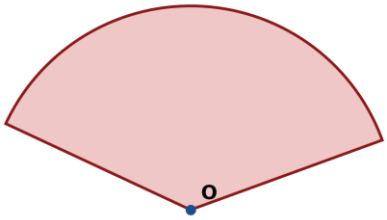
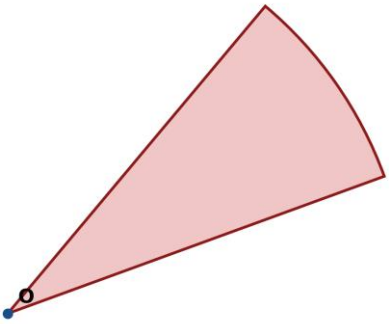
NO	SOLUTION		MARK
1 a)	$(-1-6i)+(-1+3i)$	M1 A1	4 m
	$-2-3i$	A1	
1 b)	$\frac{-4+5i}{5-3i}$	M1	5 m
	$=\frac{(-4+5i)(5+3i)}{(5-3i)(5+3i)}$	M1	
	$=\frac{-35+13i}{34}$	A1	
	$-\frac{35}{34}+\frac{13i}{34}$	A1	
1 c)	$i^{40}=i^{4(10)+0}$	M1	3 m
	$=(i^4)^{10}i^0=1^{10}i^0$	A1	
	$=1$	A1	
1 d)	$\lvert z\rvert=\sqrt{3^2+4^2}=5$	M1 A1 M1	5 m
	$\tan\theta=\frac{4}{3},\quad \theta=53.1^\circ$	A1	
2 a)	$4x-2y=10$	M1	4 m
	$x+3y=-15$	A1	
	$x=0,\quad y=-5$	A1	
2 b)	$-2x+y-3z=4$	M1	5 m
	$x-y+3z=-1$	A1	
	$-x-2y-z=0$		
	$x=-3,\quad y=1,\quad z=1$	A1	
2 c)	$8x-21=x^2-6$	M1	5 m
	$x^2-8x+15=0$	A1	
	$(x-5)(x-3)=0$		
	$x=3$ or $x=5$		
	$(3,3),\quad (5,19)$	A1	

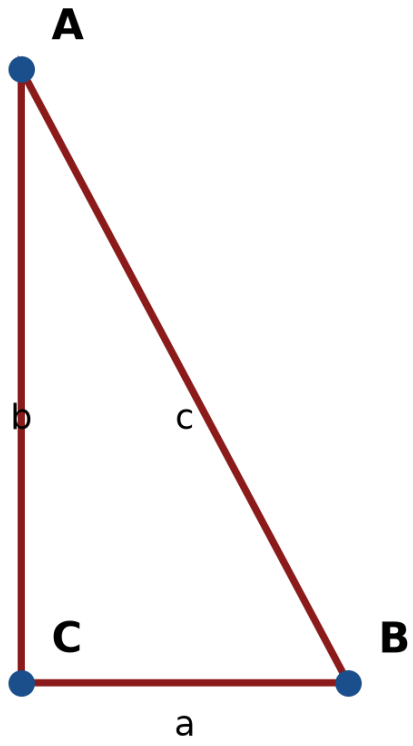
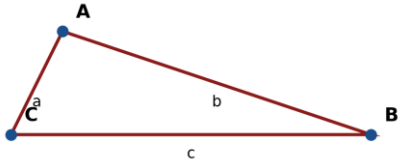
NO	SOLUTION		MARK
3 a)	$x = 0, \quad y = 0, \quad 1x + 2y = 6$ $x \geq 0, \quad y \geq 0, \quad 1x + 2y \leq 6$  	B1 B1 A1	5 m
3 b)	$\frac{x}{3} + \frac{y}{8} = 1$ $x \geq 0, \quad y \geq 0, \quad 8x + 3y \leq 24$  	M1 A1 A1	5 m
4 a)	$-4x^3 + 4x^2 - 3x + 2 = (x - 1)(-4x^2 - 3) - 1$ $q(x) = -4x^2 - 3, \quad r = -1$	M1 M1 A1 A1	5 m
4 b)	$x^3 + 5x^2 - x + 5 = (x + 3)(x^2 + 2x - 7) + 26$ $q(x) = x^2 + 2x - 7, \quad r = 26$	M1 M1 A1 A1	5 m
4 c)	$f(4) = 0$ $3x^3 - 9x^2 - 17x + 20 = (x - 4)(3x^2 + 3x - 5)$	M1 A1 M1 A1	5 m

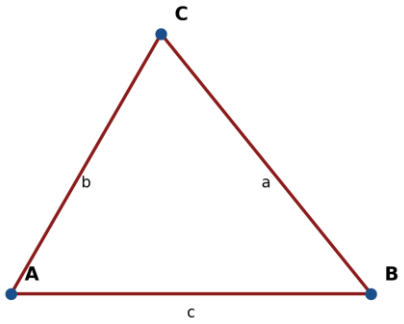
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NO	SOLUTION		MARK
4 d)	$2x^3 + 2x^2 - 5x - 2$ $f(-3) = -23$	M1 A1 A1	3 m
4 e)	$3x^3 + 6x^2 - 27x - 54$ $f(-2) = 0, \quad f(-3) = 0, \quad f(3) = 0$ $3(x-3)(x+2)(x+3)$	M1 M1 A1	5 m
5 a)	$3^{4-3x} = 3^{-x-2}$ $4 - 3x = -x - 2$ $x = 3$	M1 M1 A1	5 m
5 b)	$\log_2 [x(x-4)] = 5$ $x(x-4) = 32$ $x^2 - 4x - 32 = 0$ $x = 8$	M1 M1 M1 A1 A1	5 m
5 c)	$\log_5 (5) + \log_5 (5) = \log_5 (25)$ $\log_5 (25) = 2$	M1 A1 A1	3 m
6 a)	$2\cos^2 (x) - 2\cos (x) = 0$ $(\cos (x) - 1)\cos (x) = 0$ $\cos x = 1$ or $\cos x = 0$ $x = 0^\circ, 90^\circ, 270^\circ$	M1 A1 A1	5 m
6 b)	$\frac{\cos (x)+1}{\sin (x)} + \frac{\sin (x)}{\cos (x)+1} = \frac{2}{\sin (x)}$ $\frac{\cos (x)+1}{\sin (x)} + \frac{\sin (x)}{\cos (x)+1} = \frac{(\cos (x)+1)^2 + \sin^2 (x)}{(\cos (x)+1)\sin (x)}$ $\frac{(\cos (x)+1)^2 + \sin^2 (x)}{(\cos (x)+1)\sin (x)} = \frac{2}{\sin (x)}$ $\frac{2}{\sin (x)} = 2\csc (x)$	M1 M1 A1	5 m

NO	SOLUTION	MARK
6 c)	$\sin^2 A + \cos^2 A = 1$ M1 $\cos A = \frac{4}{5}$ A1 $\tan A = \frac{3}{4}$ $\sec A = \frac{5}{4}$ A1	4 m
6 d)	amplitude $= 3$ A1 A1 period $= \frac{360^\circ}{3} = 120^\circ$ A1 phase shift $= -\frac{0}{3} = 0^\circ$ B1 	5 m
6 e)	$\cos A = -\frac{4}{5}, \quad \sin B = -\frac{3}{5}$ M1 M1 $\sin 2A = 2\sin A\cos A = -\frac{24}{25}$ A1 $\cos(A+B) = \cos A\cos B - \sin A\sin B = -\frac{7}{25}$ A1	5 m
7 a)	$\theta = 30^\circ \times \frac{\pi}{180} = \frac{\pi}{6}$ M1 M1 $s = r\theta = 9\left(\frac{\pi}{6}\right) = \frac{3\pi}{2}$ A1 	4 m

NO	SOLUTION		MARK
7 b)	$\theta = \frac{3\pi}{4}$ $A = \frac{1}{2}r^2\theta = \frac{1}{2}(5)^2\left(\frac{3\pi}{4}\right) = \frac{75\pi}{8}$  	M1 M1 A1	4 m
7 c)	$s = r\theta = 2\pi$ $P = 2r + s = 24 + 2\pi = 2\pi + 24$  	M1 A1 A1	4 m
7 d)	$45^\circ \times \frac{\pi}{180} = \frac{\pi}{4}$ $\pi \times \frac{180}{\pi} = 180^\circ$	M1 A1 A1	3 m

NO	SOLUTION	MARK
8 a)	$8^2 + 15^2 = 17^2$ M1 $\tan A = \frac{8}{15} \rightarrow A = 28.07^\circ$ A1 $B = 90^\circ - A = 61.93^\circ$ A1 	5 m
8 b)	$\frac{a}{\sin A} = \frac{b}{\sin B}$ M1 $\frac{11}{\sin 30^\circ} = \frac{b}{\sin 45^\circ}$ M1 $b = \frac{11\sin 45^\circ}{\sin 30^\circ} = 11\sqrt{2}$ A1 	4 m

NO	SOLUTION		MARK
8 c)	$a^2 = b^2 + c^2 - 2bc\cos A$ $a^2 = 5^2 + 6^2 - 2(5)(6)\cos 120^\circ = 91$ $a = \sqrt{91}$  	M1 A1 A1	4 m
8 d)	$\text{Area} = \frac{1}{2}ab\sin C$ $\text{Area} = \frac{1}{2}(15)(6)\sin 90^\circ = 45$	M1 M1 A1	4 m
8 e)	$A_\triangle = \frac{1}{2}(10)(9) = 45$ $A_{\text{sector}} = \frac{1}{2}(4)^2\left(\frac{45\pi}{180}\right) = 2\pi$ $A_{\text{shaded}} = 45 - 2\pi = 45 - 2\pi$	M1 M1 A1	5 m